

Linear Equation

2

$$\mathbf{a} = (2,3)$$

Linear Equation

2

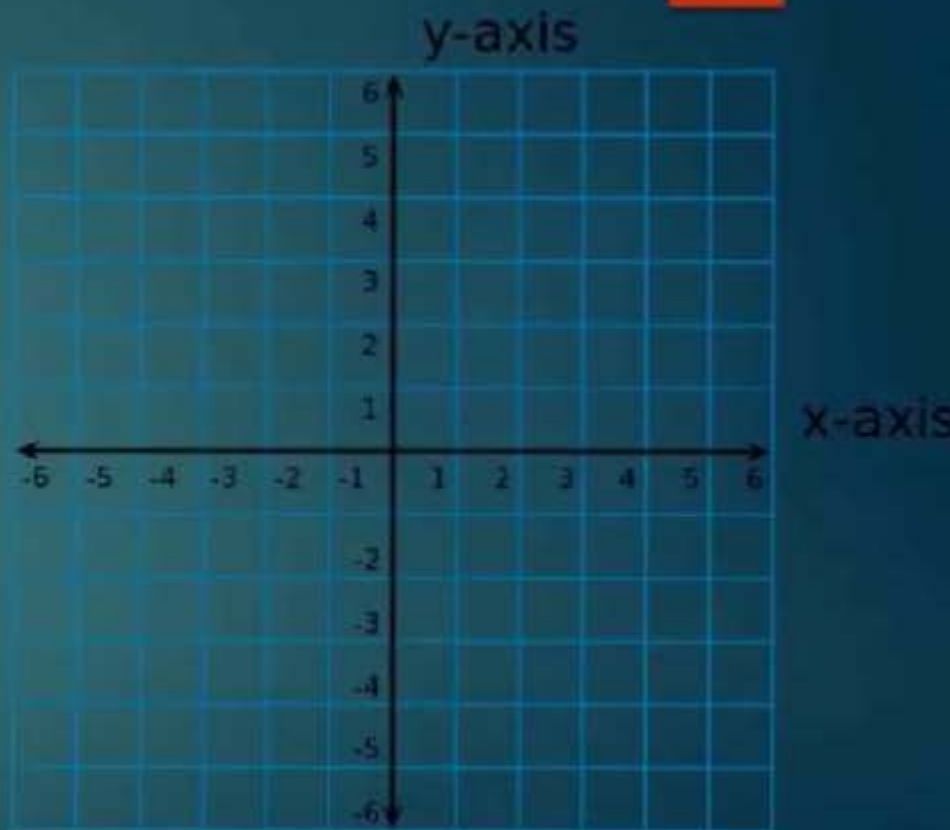
$$\mathbf{a} = (2,3)$$

$$\mathbf{b} = (-3,-2)$$

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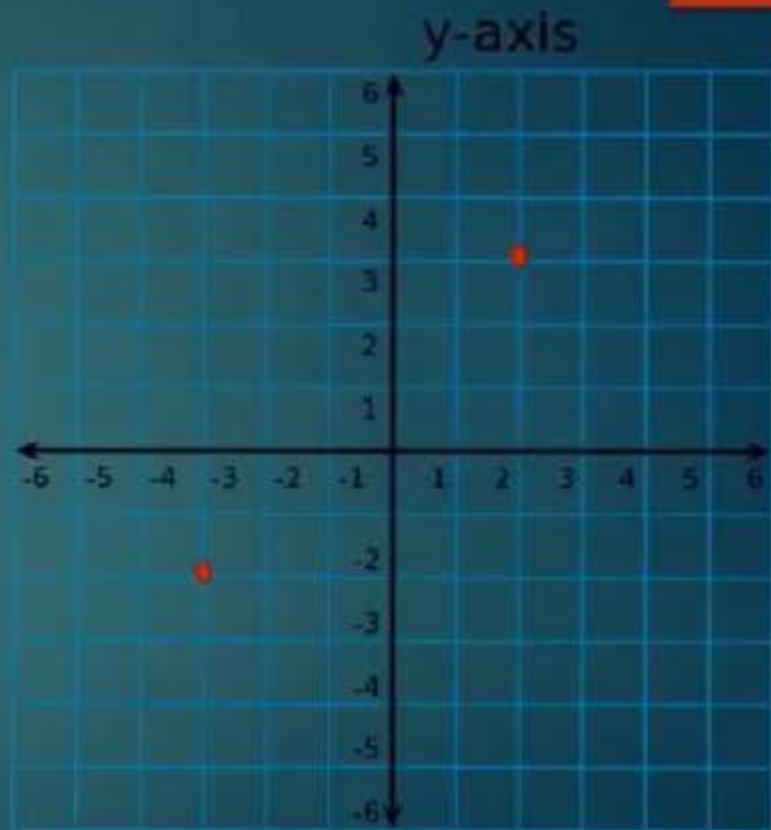


Linear Equation

$$\mathbf{a} = (2,3)$$

$$\mathbf{b} = (-3,-2)$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$



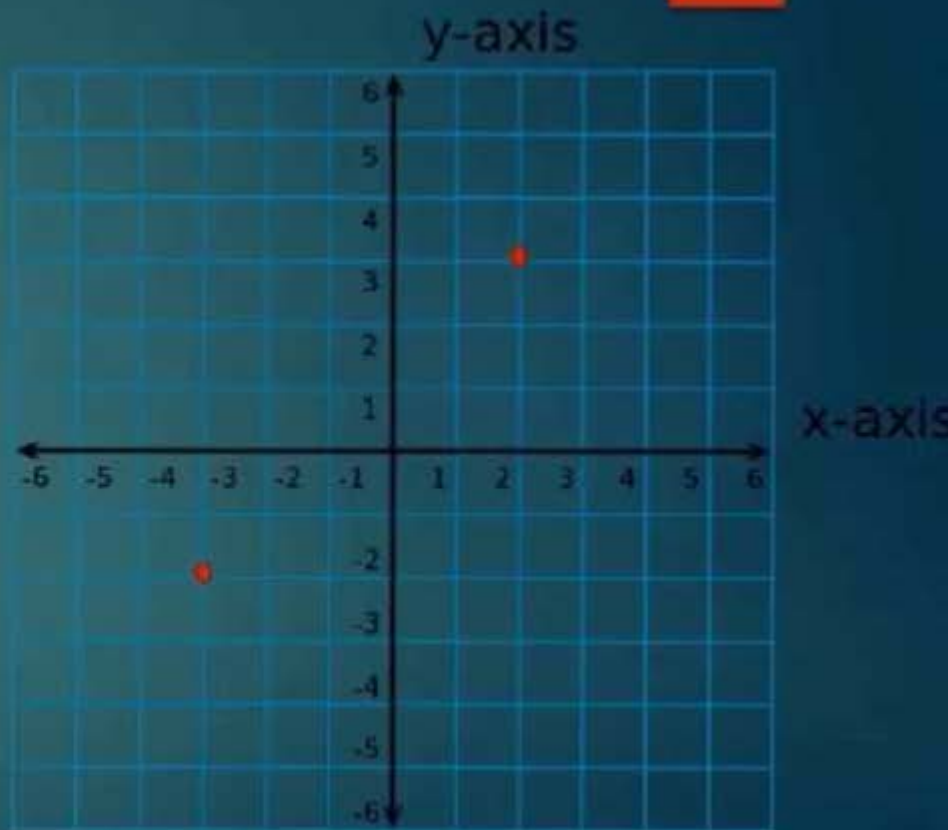
Linear Equation

$$\mathbf{a} = (2,3)$$

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$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 3}{x - 2} = \frac{-2 - 3}{-3 - 2}$$



Linear Equation

3

$$\frac{y - 3}{x - 2} = \frac{-5}{-5}$$

Linear Equation

3

$$\frac{y - 3}{x - 2} = \frac{-5}{-5}$$

$$\frac{y - 3}{x - 2} = 1$$

Linear Equation

$$\frac{y - 3}{x - 2} = \frac{-5}{-5}$$

$$\frac{y - 3}{x - 2} = 1$$

$$y - 3 = x - 2$$

Linear Equation

$$\frac{y - 3}{x - 2} = \frac{-5}{-5}$$

$$\frac{y - 3}{x - 2} = 1$$

$$y - 3 = x - 2$$

$$y = x + 1$$

Linear Equation

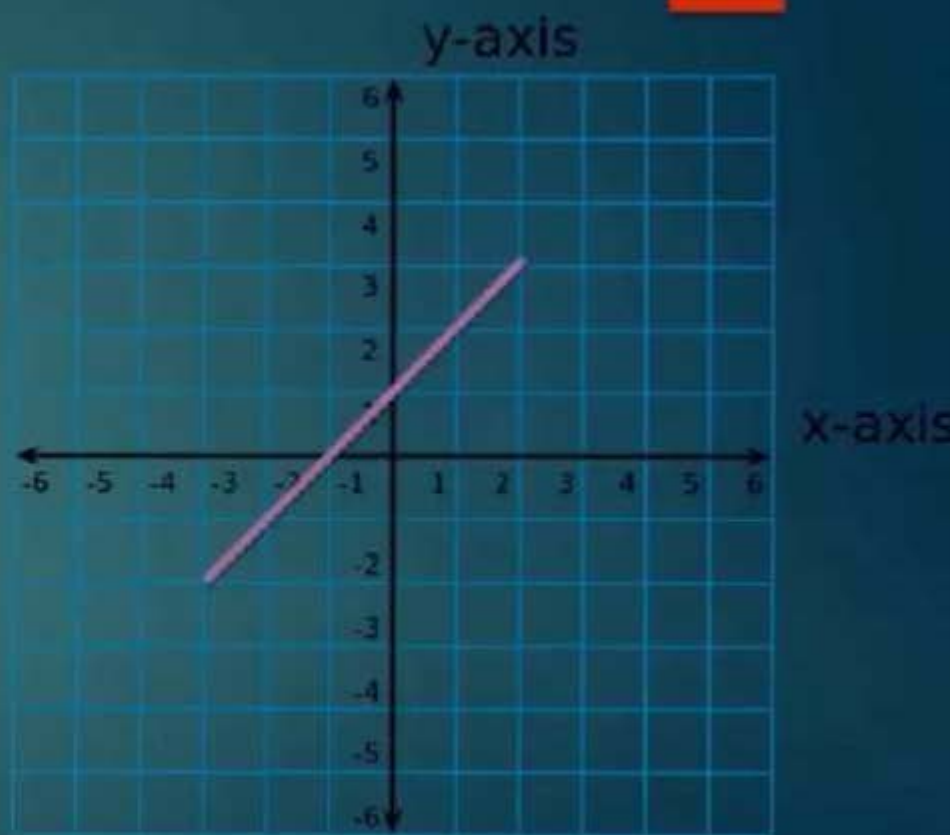
3

$$\frac{y - 3}{x - 2} = \frac{-5}{-5}$$

$$\frac{y - 3}{x - 2} = 1$$

$$y - 3 = x - 2$$

$$y = x + 1$$



Linear Equation

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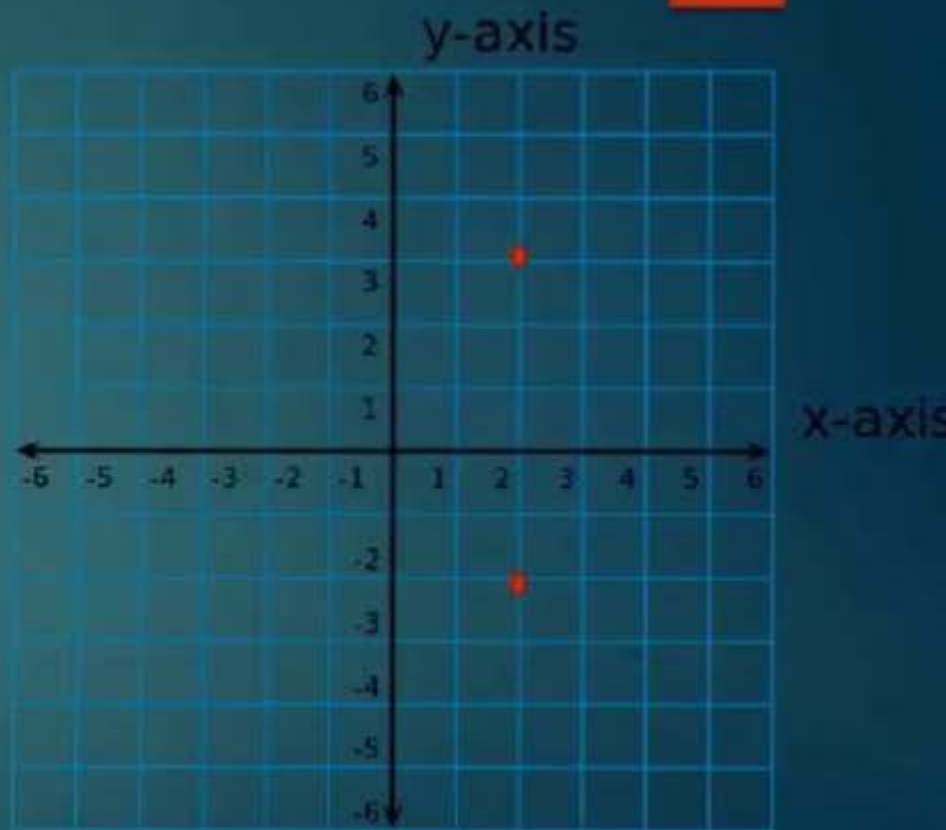
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$$\mathbf{b} = (2,-2)$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 3}{x - 2} = \frac{-2 - 3}{2 - 2}$$



Linear Equation

5

$$\frac{y - 3}{x - 2} = \frac{-5}{0}$$

Linear Equation

$$\frac{y - 3}{x - 2} = \frac{-5}{0}$$

$$0 = -5x + 10$$

Linear Equation

$$\frac{y - 3}{x - 2} = \frac{-5}{0}$$

$$0 = -5x + 10$$

$$-5x = -10$$

Linear Equation

$$\frac{y - 3}{x - 2} = \frac{-5}{0}$$

$$0 = -5x + 10$$

$$-5x = -10$$

$$x = 2$$

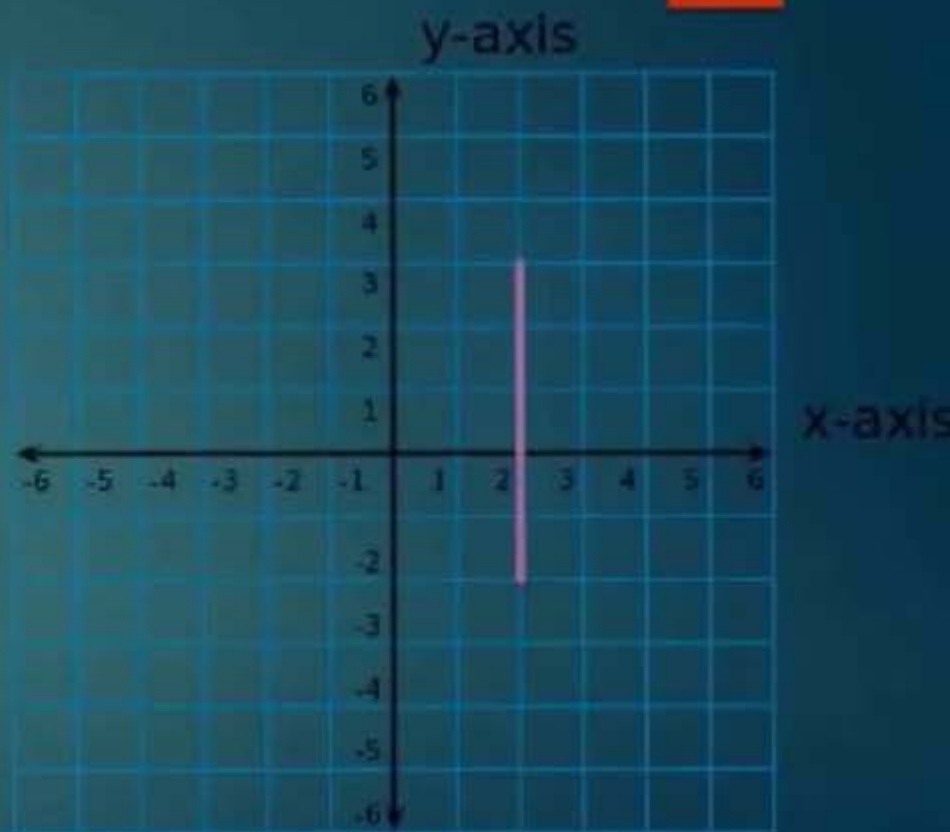
Linear Equation

$$\frac{y - 3}{x - 2} = \frac{-5}{0}$$

$$0 = -5x + 10$$

$$-5x = -10$$

$$x = 2$$



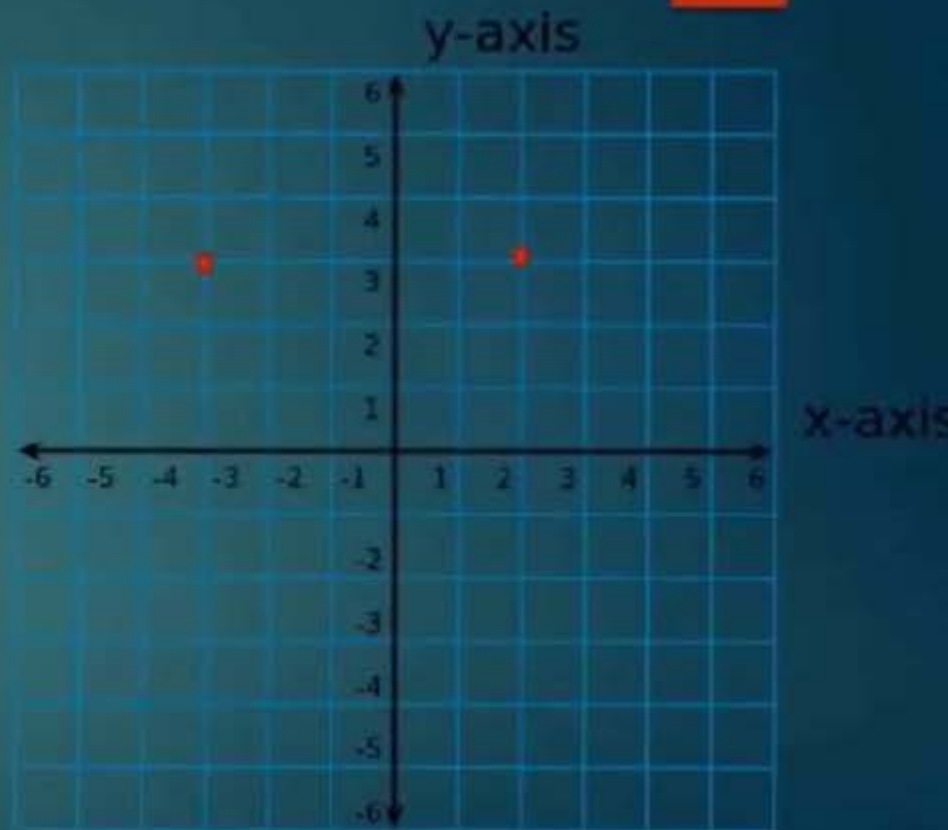
Linear Equation

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$$\mathbf{b} = (-3,3)$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 3}{x - 2} = \frac{3 - 3}{-3 - 2}$$



Linear Equation

7

$$\frac{y - 3}{x - 2} = \frac{0}{-5}$$

Linear Equation

$$\frac{y - 3}{x - 2} = \frac{0}{-5}$$

$$\frac{y - 3}{x - 2} = 0$$

Linear Equation

$$\frac{y - 3}{x - 2} = \frac{0}{-5}$$

$$\frac{y - 3}{x - 2} = 0$$

$$y - 3 = 0$$

Linear Equation

$$\frac{y - 3}{x - 2} = \frac{0}{-5}$$

$$\frac{y - 3}{x - 2} = 0$$

$$y - 3 = 0$$

$$y = 3$$

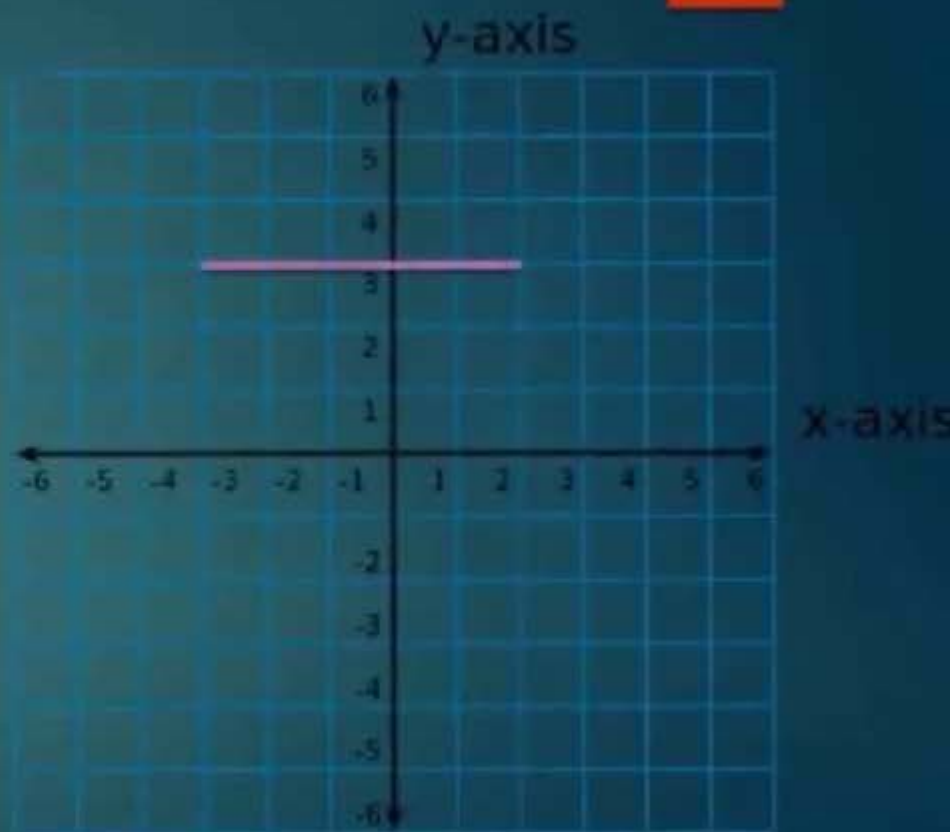
Linear Equation

$$\frac{y - 3}{x - 2} = \frac{0}{-5}$$

$$\frac{y - 3}{x - 2} = 0$$

$$y - 3 = 0$$

$$y = 3$$



Linear Equation

8

$$\mathbf{a} = (-2, 3)$$

$$\mathbf{b} = (3, -2)$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 3}{x + 2} = \frac{-2 - 3}{3 + 2}$$



Linear Equation

$$\frac{y - 3}{x + 2} = \frac{-5}{5}$$

$$\frac{y - 3}{x + 2} = -1$$

$$y - 3 = -x - 2$$

$$y = -x + 1$$

